Algebraic Geometry II

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In Algebraic Geometry I, we studied the classical approach of varieties in affine or projective space. While this is very geometric, there remain issues: Firstly, one needs the base field K to be algebraically closed: otherwise, Hilbert's Nullstellensatz fails, and the correspondence between geometric and algebraic objects breaks down (e.g. $X^2 + 1 = 0$ has no solutions in $\mathbb{R}$). In fact, we would like to not only look at non-algebraically closed fields, but more generally at algebraic sets over arbitrary rings, to study, for instance, algebraic equations over the integers using geometric methods. Lecture 1 Apr 13, 2026

Another issue of classical varieties comes from intersections: for example, take $K = \mathbb{C}$ and look at $V(y - x^2) \cap V(y) \subseteq \mathbb{A}^2$. One can compute this intersection to be $V(x^2, y) = V(x, y)$, which corresponds to the single point $(0, 0)$. However, we would like to distinguish the algebraic sets $V(x^2, y)$ and $V(x, y)$, since the former carries the extra information of "multiplicity". Algebraically, this corresponds to non-reduced K-algebras. If we had such a notion of multiplicity, we could prove more powerful theorems. For instance, Bézout's theorem simply reads $|X \cap Y| = \deg X \deg Y$, where the points in $X \cap Y$ are counted with multiplicity.

To resolve these problems, and at the same time introduce a much more general and powerful framework, Grothendieck introduced the language of schemes. In the classical setting, note that the points of an affine variety $X \subseteq \mathbb{A}^n$ correspond to the maximal ideals of $A(X)$. In the new setting, we will look at all prime ideals of $A(X)$.

For an arbitrary ring A, we can give $\operatorname{Spec} A$ a Zariski topology, and define regular functions on open subsets of $\operatorname{Spec} A$. This is all similar to the classical ideas, but involves more technical objects. This will defie the notion of an *affine scheme*.

1 Sheaves

Definition 1.1. Let X be a topological space. A *presheaf* $\mathcal{F}$ of abelian groups on X consists of an abelian group $\mathcal{F}(U)$ for each open subset $U \subseteq X$ and a group homomorphism $\rho_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ for any $V \subseteq U$ open, such that $\rho_{U,U} = \mathrm{id}_{\mathcal{F}(U)}$ and $\rho_{V,W} \circ \rho_{U,V} = \rho_{U,W}$ for any $W \subseteq V \subseteq U$ open.

In other words, a presheaf is a contravariant functor from the category of open subsets of X to the category of abelian groups. The homomorphisms $\rho_{U,V}$ are called *restriction maps*. An element $s \in \mathcal{F}(U)$ is called a section of $\mathcal{F}$ on U. If $V \subseteq U$ is open, we will usually write $s|_V := \rho_{UV}(s)$.

Remark 1.2. Analogously one defines presheaves of sets, groups, rings, R-modules etc.

Example 1.3. Let X be a topological space, and let $\mathcal{F}(U) := \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$, with ρ_{UV} the usual restriction of functions. Similarly, if X carries additional structure, one can look at conformal, differentiable or holomorphic functions on X.

Let X be a quasi-projective variety over a field K. Then the regular functions on X form a presheaf $\mathcal{O}_X(U) = \{f : U \to X \mid f \text{ regular}\}$.

In all examples, the presheaf $\mathcal{F}$ satisfies the following two properties:

Definition 1.4. A presheaf $\mathcal{F}$ on X is a *sheaf* if the following two conditions hold for any open $U \subseteq X$ and every open cover $U = \bigcup_{i \in I} U_i$:

(i) If $s, t \in \mathcal{F}(U)$ with $s|_{U_i} = t|_{U_i}$ for all $i \in I$, then $s = t$.

(ii) If $s_i \in \mathcal{F}(U_i)$ for all $i \in I$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there exists $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all $i \in I$.

In fact, the section s in (ii) is unique by (i).

Remark 1.5. The examples in 1.3 are all sheaves. For an example of a presheaf that is not necessarily a sheaf, let $1 \neq A$ be any abelian group, X a topological space, and $\mathcal{F}(U) := A$ for all opens $U \subseteq X$. If X is not connected, this is not a sheaf. To rectify this, give A the discrete topology, and define the *constant sheaf* $\underline{A}$ by

$$\underline{A}(U) := \operatorname{Hom}(U, A) = \{f : U \to A \mid f \text{ locally constant}\}$$

Definition 1.6. Let $\mathcal{F}$ be a presheaf on X, and let $x \in X$. Then the *stalk* of $\mathcal{F}$ at x is defined as the direct limit

$$\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U) = \{(U, s) \mid x \in U, s \in \mathcal{F}(U)\} / \sim,$$

where $(U, s) \sim (V, t)$ if there exists an open $x \in W \subseteq U \cap V$ with $s|_W = t|_W$.

Remark 1.7. $\mathcal{F}_x$ is an abelian group with operation $[(U, s)] + [(V, t)] = [(U \cap V, s|_{U\cap V} + t|_{U\cap V})]$.

Definition 1.8. Let X be a topological space. A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of (pre)sheaves on X is a family of morphisms $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ for all opens $U \subseteq X$, such that for all opens $V \subseteq U$, the following diagram commutes:

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\varphi_U"] \arrow[d, "\rho_{UV}"] & \mathcal{G}(U) \arrow[d, "\rho_{UV}"] \\ \mathcal{F}(V) \arrow[r, "\varphi_V"] & \mathcal{G}(V) \end{tikzcd}$$

φ is an isomorphism if there exists an inverse morphism $\psi : \mathcal{G} \to \mathcal{F}$.

Remark 1.9. In other words, a morphism of (pre)sheaves is a natural transformation of the two functors.

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Proposition 1.10. *Let $x \in X$. A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of presheves on X induces a morphism $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ of stalks, given explicitly by* $[(U, f)] \mapsto [(U, \varphi(U)(f))]$.

Proof. This follows directly from the universal property of direct limits. □

Proposition 1.11. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of schemes. Then φ is an isomorphism if and only if $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is an isomorphism for every $x \in X$.*

Note that this fails for presheaves, and hence gives some motivation for the conditions in the definition of a sheaf.

Proof. If φ is an isomorphism, then so is φ_x (more precisely $(\varphi_x)^{-1} = (\varphi_x)^{-1}$) by functoriality of direct limits.

Conversely, assume that φ_x is an isomorphism for every $x \in X$, and let $U \subseteq X$ be open. Assume $s \in \ker \varphi_U$. For any $x \in U$, we have $\varphi_x([(U, s)]) = [U, \varphi(U)(s)] = 0$, so by injectivity of φ_x we get $[(U, s)] = 0$ in $\mathcal{F}_x$. That is, $s = 0$ on some neighbourhood $x \in U_x \subseteq U$. Then $U = \bigcup_{x\in U} U_x$, so by 1.4(1) we get $s = 0 \in \mathcal{F}(U)$. Hence φ_U is injective.

For surjectivity, let $t \in \mathcal{G}(U)$. For any $x \in U$, φ_x is surjective, so let $[(U_x, s_x)] \in \mathcal{F}_x$ be a preimage of $[(U, t)] \in \mathcal{G}_x$, wlog $U_x \subseteq U$. Then $U = \bigcup_{x\in X} U_x$ and $s_x \in \mathcal{F}(U_x)$. To check the gluing condition, let $x, y \in U$ and compute

$$\varphi_{U_x\cap U_y}(s_x|_{U_x\cap U_y}) \overset{1.8}{=} (\varphi_{U_x}(s_x))|_{U_x\cap U_y} = t|_{U_x}|_{U_x\cap U_y} \overset{1.1}{=} t|_{U_x\cap U_y} = \varphi_{U_x\cap U_y}(s_y|_{U_x\cap U_y}),$$

thus by injectivity $s_x|_{U_x \cap U_y} = s_y|_{U_x \cap U_y}$. Hence by 1.4(2) we obtain an element $s \in \mathcal{F}(U)$ s.t. $s|_{U_x} = s_x$ for every $x \in X$. It remains to check that $\varphi_U(s) = t$. But this holds by 1.4(1) applied to $\mathcal{G}$, since by construction $\varphi_U(s)|_{U_x} = \varphi_{U_x}(s|_{U_x}) = \varphi_{U_x}(s_x) = t|_{U_x}$. Therefore, φ_U is surjective, hence an isomorphism. □

Definition 1.12 (Naive[1])**.** Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. We say that φ is injective/surjective if for every $x \in X$, $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective/surjective. In particular, if $\mathcal{F}, \mathcal{G}$ are sheaves, then by proposition 1.11, φ is an isomorphism if and only if it is injective and surjective.

Definition 1.13. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves on X. We define the following presheaves:

$$\begin{aligned} \ker(\varphi)(U) &:= \ker(\varphi_U), \\ \operatorname{im}(\varphi)^{\text{pre}}(U) &:= \operatorname{im}(\varphi_U) \\ \operatorname{coker}(\varphi)^{\text{pre}}(U) &:= \operatorname{coker}(\varphi_U) \end{aligned}$$

Remark 1.14. In each case, one easily checks that these assignments define presheaves. If $\mathcal{F}, \mathcal{G}$ are sheaves, then $\ker \varphi$ is a sheaf as well (Exercise), but in general $\operatorname{im} \varphi$ and $\operatorname{coker} \varphi$ are not sheaves.

Proposition 1.15 (Sheafification)**.** *Let $\mathcal{F}$ be a presheaf on X. Then there is a unique (up to unique isomorphism) sheaf $\mathcal{F}^+$, called* sheafification of $\mathcal{F}$*, and a morphism $\theta : \mathcal{F} \to \mathcal{F}^+$ with the following universal property: For any sheaf $\mathcal{G}$ and morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of presheaves, there exists a unique morphism $\widetilde{\varphi} : \mathcal{F}^+ \to \mathcal{G}$ making the following diagram commute:*

$$\begin{tikzcd} \mathcal{F} \arrow[rr, "\varphi"] \arrow[dr, "\theta"'] & & \mathcal{G} \\ & \mathcal{F}^+ \arrow[ur, dashed, "\widetilde{\varphi}"'] & \end{tikzcd}$$

Moreover, θ induces isomorphisms $\theta_x : \mathcal{F}_x \to \mathcal{F}^+_x$ for every $x \in X$. If $\mathcal{F}$ is already a sheaf, then θ is an isomorphism.

Proof. We construct $\mathcal{F}^+$ as follows:

$$\mathcal{F}^+(U) := \left\{ s : U \to \coprod_{x \in U} \mathcal{F}_x \mid s(x) \in \mathcal{F}_x \text{ for all } x \in U, \text{ and } (*) \right\},$$

where $(*)$ is the following extra condition: For every $x \in U$ there exists an open neighbourhood $x \in V \subseteq U$ and $t \in \mathcal{F}(V)$ with $s(y) = t_y := [(V, t)]$ for all $y \in V$. Further, set $\theta_U(t)(x) = t_x$ for all $U \subseteq X$ open, $t \in \mathcal{F}(U)$ and $x \in U$. One checks that this $\mathcal{F}^+, \theta$ have the required properties:

It is clear that $\mathcal{F}^+$ is a sheaf, since it is constructed via functions with local properties, and that Lecture 3
θ is a morphism of presheaves, since $(t|_W)_x = t_x$ for $x \in W \subseteq V$, $t \in \mathcal{F}(V)$. Let $x \in X$ and Apr 20, 2026
$[(V, t)] \in \ker \theta_x$. Up to shrinking V, we have $\theta_V(t) = 0$. In particular, $[(V, t)] = \theta_V(t)(x) = 0$. Let $[(V, s)] \in \mathcal{F}^+_x$. By $(*)$ wlog there is $t \in \mathcal{F}(V)$ s.t. $s(y) = t_y$ for all $y \in V$. Hence $\theta_V(t) = [(V, s)]$. If $\mathcal{F}$ is a sheaf, then θ is thus an isomorphism by proposition 1.11.

Finally, we check the universal property: Let $\mathcal{G}$ be a sheaf, and $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. By the previous paragraph, it suffices to give a morphism $\widetilde{\varphi} : \mathcal{F}^+ \to \mathcal{G}^+ \cong \mathcal{G}$. Let $U \subseteq X$ be open and let $\widetilde{\varphi}(s)(y) = \varphi_y(s(y))$. It is easy to check that this is a well-defined morphism of sheaves, makes the diagram commute, and is the only morphism with this property. The details are left as an exercise. □

[1] will be corrected in 1.21

Example 1.16. Let X be a topological space such that for every $x \in X$ there is a neighbourhood $x \in U_x \subsetneq X$, and let $0 \neq A$ an abelian group.

(i) Let $\mathcal{F}(X) = A$ and $\mathcal{F}(U) = 0$ otherwise. This is clearly not a sheaf. By the assumption on X we have $\mathcal{F}_x = 0$ for all $x \in X$, so $\mathcal{F}^+ = 0$.
(ii) Let $\mathcal{F}$ be the constant presheaf with value A as in remark 1.5. From the universal property one sees that $\mathcal{F}^+ = \underline{A}$ is the constant sheaf.

Definition 1.17. A sub-(pre)sheaf of a (pre)sheaf $\mathcal{F}$ on X is a (pre)sheaf $\mathcal{F}'$ s.t. for every open $U \subseteq X$ one has $\mathcal{F}'(U) \subseteq \mathcal{F}(U)$, and for every $V \subseteq U$ open, $\rho'_{UV} = \rho_{UV}|_{\mathcal{F}'(U)}$. In other words, the inclusions $\mathcal{F}'(U) \hookrightarrow \mathcal{F}(U)$ have to form a morphism of (pre)sheaves.

If $\mathcal{F}' \subseteq \mathcal{F}$ is a sub-presheaf, then for every $x \in X$ we have $\mathcal{F}'_x \subseteq \mathcal{F}_x$.

Proposition 1.18. *Let $\mathcal{F}$ be a sheaf and $\mathcal{F}'$ be a subsheaf. Then $\mathcal{F}' = \mathcal{F}$ if and only if $\mathcal{F}'_x = \mathcal{F}_x$ for all $x \in X$.*

Proof. Follows directly from proposition 1.11. □

Example 1.19. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves, then $\ker\varphi$ and $\operatorname{im}\varphi^{\text{pre}}$ are sub-presheaves of $\mathcal{F}$ resp. $\mathcal{G}$.

Exercise 1.20. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. Show that $(\ker\varphi)_x = \ker(\varphi_x)$ and $(\operatorname{im}\varphi^{\text{pre}})_x = \operatorname{im}(\varphi_x)$.

Definition 1.21. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. We say that φ is injective if and only if $\ker\varphi = 0$. For presheaves, we say φ is surjective if $\operatorname{coker}\varphi^{\text{pre}} = 0$[2] If $\mathcal{F}, \mathcal{G}$ are sheaves, define the image and cokernel sheaves via $\operatorname{im}\varphi := (\operatorname{im}\varphi^{\text{pre}})^+$ and $\operatorname{coker}\varphi := (\operatorname{coker}\varphi^{\text{pre}})^+$. Then $\operatorname{im}\varphi$ can be identified with a subsheaf of $\mathcal{G}$ (see next proposition). We say that φ is surjective as a morphism of sheaves if $\operatorname{im}\varphi = G$, equivalently $\operatorname{coker}\varphi = 0$.

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Proposition 1.22. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then $\operatorname{im}\varphi$ can be seen as a subsheaf of $\mathcal{G}$. Moreover, $\operatorname{im}\varphi = \mathcal{G} \iff \varphi_x$ is surjective for all $x \in X$.*

Proof. We have $\operatorname{im}\varphi \subseteq \mathcal{G}^+ \cong \mathcal{G}$. The second part follows directly from proposition 1.11. □

Remark 1.23. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of (pre)sheaves. Then φ is injective if and only if φ_U is injective for all $U \subseteq X$ open, that is $\ker\varphi = 0$. If φ is injective, then also $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective, but the converse is only true for sheaves.

φ is a surjective morphism of presheaves if φ_U is surjective for all $U \subseteq X$ open, that is $\operatorname{im}\varphi^{\text{pre}} = \mathcal{G}$. Again, if φ is surjective, then so are all φ_x, but the converse is not true in general (even for sheaves).

If $\mathcal{F}, \mathcal{G}$ are sheaves, φ is a surjective morphism of sheaves if and only if $\operatorname{im}\varphi = \mathcal{G}$, if and only if φ_x is surjective for all $x \in X$.

Example 1.24. (i) Let $\mathcal{F}, \mathcal{G}$ be (pre)sheaves on X. Then $(F \oplus \mathcal{G})(U) := \mathcal{F}(U) \oplus \mathcal{G}(U)$ is again a (pre)sheaf.
(ii) Let A be an abelian group, X a topological space and $x \in X$. The sheaf

$$ {}^x\mathcal{F}(U) := \begin{cases} A, & \text{if } x \in U, \\ 0 & \text{otherwise} \end{cases} $$

is called the *skyscraper sheaf*. One calculates ${}^x\mathcal{F}_y = A$ if $y \in \overline{\{x\}}$, and ${}^x\mathcal{F}_y = 0$ otherwise. In particular, if X is T_1 (that is, points are closed), then ${}^x\mathcal{F}_y \neq 0$ only for $x = y$.

[2] Note that this is not equivalent to φ_X being surjective for all $x \in X$.

(iii) Let X be a connected T_1-space, $x \neq z \in X$. Let $0 \neq A$ be an abelian group, and consider the natural map $\varphi : \underline{A} \to {}^x\mathcal{F} \oplus {}^z\mathcal{F}$. One sees that φ is surjective as a morphism of sheaves, but not as a morphism of presheaves.

Definition 1.25. Let $f : X \to Y$ be a continuous map of topological spaces.

(i) Let $\mathcal{F}$ be a sheaf on X. The *direct image* of $\mathcal{F}$ is the sheaf $f_*\mathcal{F}$ on Y defined by

$$f_*\mathcal{F}(U) := \mathcal{F}(f^{-1}(U)).$$

(ii) Let $\mathcal{G}$ be a sheaf on Y. The *inverse image* $f^{-1}\mathcal{G}$ of $\mathcal{G}$ is defined as the sheafification of the presheaf

$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V).$$

2 Affine Schemes

In Algebraic Geometry I, we showed an equivalence of categories between affine algebraic sets over K, and finitely generated reduced K-algebras, for K algebraically closed. We want to generalize this to allow any ring A on the algebraic side. A natural generalization of the classical setting is to consider $\operatorname{Spec} A$, and to endow this set with topological and algebraic information.

Let A be a ring (always commutative with 1).

Definition 2.1. $\operatorname{Spec} A := \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ prime ideal}\}$. If $I \subseteq A$ is any ideal, let $V(I) := \{\mathfrak{p} \in \operatorname{Spec} A \mid I \subseteq \mathfrak{p}\}$. (Note that there is a canonical bijection $V(I) \cong \operatorname{Spec} A/I$.) For $S \subseteq A$ any subset, we write $V(S) := V(\langle S \rangle)$.

Lemma 2.2. *Let $I, J \subseteq A$ be ideals.*

(i) $\sqrt{I} := \{f \in A \mid f^n \in I \text{ for some } n \geq 1\} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}$.
(ii) $I \subseteq \sqrt{I}$ *and* $V(I) = V(\sqrt{I})$.
(iii) If $I \subseteq J$, *then* $V(J) \subseteq V(I)$.
(iv) If $V(J) \subseteq V(I)$ *if and only if* $\sqrt{I} \subseteq \sqrt{J}$.

Proof. (i) was proven in Commutative Algebra. (ii), (iii), and one direction of (iv) are clear. So assume $V(J) \subseteq V(I)$, then $\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \subseteq \bigcap_{\mathfrak{p} \in V(J)} = \sqrt{J}$ by (i). □

Lemma 2.3. *Let $I, J, \{I_\alpha\}_{\alpha \in I} \subseteq A$ be ideals.*

(i) $V(IJ) = V(I \cap J) = V(I) \cup V(J)$.
(ii) $V(\sum_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$.

Proof. (i) For a prime ideal $\mathfrak{p} \subseteq A$, we have $IJ \subseteq \mathfrak{p}$ if and only if $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$, if and only if $I \cap J \subseteq \mathfrak{p}$.

(ii) Exercise. □

Definition 2.4. By lemma 2.3 (and the observations $V(A) = \emptyset$, $V(0) = \operatorname{Spec} A$), the collection $\{V(I) \mid I \subseteq A \text{ ideal}\}$ forms the set of closed sets of a topology on $\operatorname{Spec} A$, the so-called *Zariski topology*.

Remark 2.5. Not every point of $\operatorname{Spec} A$ is closed. More precisely, let $x \in \operatorname{Spec} A$. Then $\overline{\{x\}} = V(x)$, so $\{x\}$ is closed if and only if x is a maximal ideal.

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Example 2.6. (i) If A is an integral domain, then $0 \in \operatorname{Spec} A$ satisfies $\overline{\{0\}} = \operatorname{Spec} A$. In particular, every open in $\operatorname{Spec} A$ contains $\{0\}$. Such a point is called a *generic point* of $\operatorname{Spec} A$.

(ii) Let $\mathbb{A}^1 := \operatorname{Spec} K[x]$. Then $\mathbb{A}^1$ contains the generic point (0), plus closed points corresponding to normalised irreducible polynomials in $K[x]$. If K is algebraically closed, every irreducible polynomial is linear, so $\mathbb{A}^1$ looks very similar to affine space in the classical setting, the only difference being the additional generic point.

(iii) $\operatorname{Spec} \mathbb{Z} = \{(0)\} \cup \{(p) \mid p \text{ prime }\}$

(iv) Let $\mathbb{A}^2 := \operatorname{Spec} K[x, y]$ with K algebraically closed. It contains the generic point (0), closed points $(x - a, y - b)$ as in the classical settings, but also (principal) prime ideals corresponding to curves. Let (F) be such an ideal, then

$$\overline{\{(F)\}} = \{(F)\} \cup \{(x - a, y - b) \mid F(a, b) = 0\}.$$

Lemma 2.7. *(i) If A is an integral domain, then* $\operatorname{Spec} A$ *is irreducible. (The converse fails e.g. for $K[x]/(x^2)$). If A is reduced and* $\operatorname{Spec} A$ *is irreducible, then A is an integral domain.*

(ii) $\operatorname{Spec} A$ *is not connected if and only if there exists* $0, 1 \neq e \in A$ *with* $e^2 = e$*, if and only if* $A \cong A_1 \times A_2$ *with* $A_1, A_2 \neq 0$.

Proof. (i) Assume that $\operatorname{Spec} A$ is irreducible and A is reduced but not an integral domain. Let $a * b = 0$ with $0 \neq a, b \in A$. Then $\operatorname{Spec} A = V(0) = V(a) \cup V(b)$. If $V(a) = \operatorname{Spec} A$, then $a \in \sqrt{0} = 0$ by assumption, contradiction. Hence $\operatorname{Spec} A$ is reducible. The rest is left as an exercise. □

Definition 2.8. Let $f \in A$. Then $D(f) := \operatorname{Spec} A \setminus V(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}$ is called a *distinguished* or *standard open* subset of $\operatorname{Spec} A$.

Remark 2.9. One has $D(f) \cap D(g) = D(fg)$. The set $\{D(f) \mid f \in A\}$ forms a basis of the Zariski topology on $\operatorname{Spec} A$: Let $U = \operatorname{Spec} A \setminus V(I)$ be open and $\mathfrak{p} \in U$. Since $\mathfrak{p} \notin V(I)$, $I \not\subseteq \mathfrak{p}$. Hence choose $f \in I \setminus \mathfrak{p}$, which satisfies $\mathfrak{p} \in D(f) \subseteq U$.

We want to construct a sheaf of "regular functions" on $\operatorname{Spec} A$. Recall from the classical setting, that if $X \subseteq \mathbb{A}^n$ was an affine algebraic set and $U \subseteq X$ open, we defined $\mathcal{O}_X(U)$ as the set of all functions $f : U \to K$ that locally are rational functions, and we proved $\mathcal{O}_X(D(f)) \cong A(X)_f$ and $\mathcal{O}_{X,x} \cong \varinjlim_{D(f) \ni x} A(X)_f \cong A(X)_{\mathfrak{m}_x}$. This motivates us to define a sheaf $\mathcal{O}_X$ of rings on $\operatorname{Spec} A$ s.t. $\mathcal{O}_X(D(f)) \cong A_f$. Note that if $D(g) \subseteq D(f)$, then $g^m = fa$ for some $a \in A, m \geq 1$ by lemma 2.2, so there is a natural map $A_f \to A_g$. Since we know what the stalks are supposed to look like, we can proceed as in the proof of sheafification 1.15.

Definition 2.10. Let $X = \operatorname{Spec} A$. The structure sheaf $\mathcal{O}_X$ is defined as

$$\mathcal{O}_X(U) := \left\{ s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}} \mid s(\mathfrak{p}) \in A_{\mathfrak{p}} \text{ and } (*) \right\},$$

where $(*)$ is the extra condition that for every $\mathfrak{p} \in U$ there exists an open neighbourhood $\mathfrak{p} \in V \subseteq U$ and $a, f \in A$ such that $s(\mathfrak{q}) = \frac{a}{f}$ in $A_{\mathfrak{q}}$ for every $\mathfrak{q} \in V$. By construction it is clear that $\mathcal{O}_X$ is a sheaf of rings on $\operatorname{Spec} A$.

Proposition 2.11. *For* $\mathfrak{p} \in \operatorname{Spec} A$ *we have* $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$*, and for* $f \in A$*,* $\mathcal{O}_X(D(f)) \cong A_f$*. In particular,* $\mathcal{O}_X(X) \cong A$.

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Proof. Let $\mathfrak{p} \in \operatorname{Spec} A$ and set $\varphi : \mathcal{O}_{X,\mathfrak{p}} \to A_{\mathfrak{p}}$, $[(U, f)] \mapsto f(\mathfrak{p})$. This is a well-defined homomorphism of rings. Let $\frac{a}{s} \in A_{\mathfrak{p}}$ with $a, s \in A$, $s \notin \mathfrak{p}$. That is, $\mathfrak{p} \in D(s)$, so $[(D(s), \mathfrak{q} \mapsto \frac{a}{s})]$ is a well-defined preimage of $\frac{a}{s}$. Conversely, let $[(U, f)] \in \ker \varphi$. By shrinking U, we may assume $f(\mathfrak{q}) = \frac{a}{s} \in A_{\mathfrak{q}}$, $a, s \in A$, for all $\mathfrak{q} \in U$. Then $0 = \varphi([(U, f)]) = \frac{a}{s} \in A_{\mathfrak{p}}$, hence $s'a = 0$ for some $s' \notin \mathfrak{p}$, i.e. $\rho \in D(s')$. By shrinking U further, if necessary, we may assume $U \subseteq D(s')$. Then $f = \frac{a}{s} = \frac{s'a}{s's} = 0$. Therefore, φ is an isomorphism $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$.

Now let $f \in A$. Define $\psi : A_f \to \mathcal{O}_X(D(f))$, $\alpha \mapsto (t_\alpha : \mathfrak{q} \mapsto \alpha)$. This is clearly a well-defined homomorphism of rings. Let $\alpha = \frac{a}{f^n} \in \ker \psi$. That is, $\frac{a}{f^n} = 0$ in $A_{\mathfrak{q}}$ for every $\mathfrak{q} \in D(f)$. This means that for every $\mathfrak{q} \in D(f)$ there exists $s_{\mathfrak{q}} \notin \mathfrak{q}$ with $s_{\mathfrak{q}} a = 0$. Consider the ideal $I = \{b \in a \mid ba = 0\}$ (called the annihilator of a). Then for every $\mathfrak{q} \in D(f)$, $s_{\mathfrak{q}} \in I \setminus \mathfrak{q}$, i.e. $I \not\subseteq \mathfrak{q}$. Hence $V(I) \subseteq \operatorname{Spec} A \setminus D(f) = V(f)$. By lemma 2.2 it follows that $f^m \in I$. for some $m > 0$. Then $\alpha = \frac{a}{f^n} = \frac{af^m}{f^{n+m}} = 0$.

For surjectivity, let $t \in \mathcal{O}_X(D(f))$. By definition, there is an open cover $D(f) = \bigcup_i V_i$ s.t. $t|_{V_i} = \frac{a_i}{s_i}$ for some $a_i, s_i \in A$ with $V_i \subseteq D(s_i)$. By remark 2.9 wlog $V_i = D(h_i)$. Now $V(h_i) \supseteq V(s_i)$, so again by lemma 2.2, $h_i \in \sqrt{(s_i)}$, i.e. $h_i^{n_i} = s_i c_i$ for some $n_i > 0, c_i \in A$. It follows that $t|_{D(h_i)} = \frac{a_i}{s_i} = \frac{a_i c_i}{h_i^{n_i}}$. Replacing $h_i^{n_i}$ by h_i (since $D(h_i) = D(h_i^{n_i})$) and $a_i c_i$ by a_i, we simply write $t|_{D(h_i)} = \frac{a_i}{h_i}$.

Observe that there exists a finite subcover $D(f) = \bigcup_{i=1}^k D(h_i)$ of $\bigcup_i D(h_i)$. Indeed, $D(f) \subseteq \bigcup_i D(h_i)$ implies $V(f) \supseteq \bigcap_i V(h_i) = V(\sum_i (h_i))$ by lemma 2.3. By lemma 2.2, $f^n \in \sum_i (h_i)$ for some $n > 0$. But this means f^n is a *finite* linear combination of the h_i's. Taking only those h_i which appear in this sum and retracing the argument backwards yields the claim.

For $i \neq j$, we have $\frac{a_i h_j}{h_i h_j} = t|_{D(h_i)}|_{D(h_i h_j)} = t|_{D(h_j)}|_{D(h_i h_j)} = \frac{a_j h_i}{h_j h_i}$ as functions on $D(h_i h_j)$. By the injectivity of ψ (for $D(h_i h_j)$) proved above, even $\frac{a_i h_j}{h_i h_j} = \frac{a_j h_i}{h_j h_i}$ as elements of $A_{h_i h_j}$, i.e. $(h_i h_j)^n (a_i h_j - a_j h_i) = 0$ in A, for some n. Since there are only finitely many sets in the open cover, we can choose n big enough such that this equality holds for all i, j. Lecture 7 May 4, 2026

Replacing h_i by h_i^{n+1} and a_i by $h_i^n a_i$, we still have $t|_{D(h_i)} = \frac{a_i}{h_i}$, and now $h_j a_i = h_i a_j$ for all i, j. As before, $f \in \sqrt{(h_1, \dots, h_k)}$, so we can write $f^m = \sum_i b_i h_i$ for some $m > 0$ and $b_i \in A$. Let $a := \sum_i b_i a_i$. We claim that $\psi(\frac{a}{f^m}) = t$. Indeed, for every j we have $f^m a_j = \sum_i b_i h_i a_j = \sum_i b_i a_i h_j = a h_j$, i.e. $\frac{a}{f^m} = \frac{a_j}{h_j} \in A_{\mathfrak{q}}$ for every $\mathfrak{q} \in D(h_j)$. □

Having now associated to a ring A the pair $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$, we next want to define morphisms between these objects. We let us motivate by the following

Remark 2.12. Recall from classical algebraic geometry that if X, Y are quasi-projective algebraic sets, a map $\varphi : X \to Y$ was defined to be a morphism if φ is continuous, and for every open $U \subseteq Y$ and $f \in \mathcal{O}_Y(U)$, its preimage $\varphi^* f = f \circ \varphi$ is regular, i.e. lies in $\mathcal{O}_X(\varphi^{-1}(U))$. This yielded compatible homomorphisms of K-algebras $\mathcal{O}_Y(U) \to \mathcal{O}_X(\varphi^{-1}(U))$, in other words a morphism of sheaves $\mathcal{O}_Y \to \varphi_* \mathcal{O}_X$.

Definition 2.13.
- (i) A *ringed space* is a pair $(X, \mathcal{O}_X)$, where X is a topological space and $\mathcal{O}_X$ a sheaf of rings on X.
- (ii) A morphism of ringed spaces $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ is a pair $(f, f^\sharp)$, where $f : X \to Y$ is a continuous map, and $f^\sharp : \mathcal{O}_Y \to f_* \mathcal{O}_X$ is a morphism of sheaves.
- (iii) A *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that all stalks $\mathcal{O}_{X,x}$, $x \in X$ are local rings.
- (iv) A morphism of locally ringed spaces $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces such that for every $x \in X$, the map $f_x^\sharp : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is a local homomorphism of rings (where a morphism of rings $\varphi : (A, \mathfrak{m}) \to (B, \mathfrak{n})$ of local rings is called *local* if $\varphi^{-1}(\mathfrak{n}) = \mathfrak{m}$). The map $f_x^\sharp$ is defined as the compositum

$$\mathcal{O}_{Y,f(x)} \xrightarrow{f^\sharp_{f(x)}} (f_* \mathcal{O}_X)_{f(x)} = \varinjlim_{f^{-1}(U) \ni x} \mathcal{O}_X(f^{-1}(U)) \xrightarrow{\text{incl}_*} \varinjlim_{U \ni x} \mathcal{O}_X(U) = \mathcal{O}_{X,x},$$

where the second map is induced by inclusion of directed sets.

Remark 2.14. Let A be a ring. The pair $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ is a locally ringed space by proposition 2.11.

Definition 2.15.
- (i) Let $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ and $(g, g^\sharp) : (Y, \mathcal{O}_Y) \to (Z, \mathcal{O}_Z)$ be morphisms of (locally) ringed spaces. Then their composition is defined as $(g, g^\sharp) \circ (f, f^\sharp) := (g \circ f, (g \circ f)^\sharp)$, with $(g \circ f)^\sharp := g_* f^\sharp \circ g^\sharp : \mathcal{O}_Z \to (g \circ f)_* \mathcal{O}_X$.
- (ii) An isomorphism of (locally) ringed spaces is a morphism with a two-sided inverse morphism.

A morphism $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is an isomorphism if and only if $f : X \to Y$ is a homeomorphism and $f^\sharp : \mathcal{O}_Y \to f_* \mathcal{O}_X$ is an isomorphism of sheaves. (Exercise)

Lemma 2.16. *Let $\varphi : A \to B$ be a morphism of rings. Then $f : \operatorname{Spec} B \to \operatorname{Spec} A$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is continuous.*

Proof. f is well-defined since the preimage of a prime ideal is again prime. Let $V(I) \subseteq \operatorname{Spec} A$. Then $f^{-1}(V(I)) = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi^{-1}(\mathfrak{p}) \in V(I)\} = V(\varphi(I))$ is closed. $\square$

Remark 2.17. Let $\varphi : A \to B$ be a ring morphism and $X = \operatorname{Spec} B, Y = \operatorname{Spec} A$. Then φ induces a morphism of locally ringed spaces $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, where f is the map of lemma 2.16, and $f^\sharp : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is defined on $U \subseteq Y$ as Lecture 8 May 7, 2026

$$f^\sharp_U : \mathcal{O}_Y(U) \to \mathcal{O}_X(f^{-1}(U)), \quad t \mapsto (\mathfrak{p} \mapsto \varphi_\mathfrak{p}(t(f(\mathfrak{p})))),$$

where $\varphi_\mathfrak{p} : A_{f(\mathfrak{p})} \to B_\mathfrak{p}$ is the natural morphism induced by the universal property of localisation. Note that if $t|_V = \frac{a}{s}$, then $f^\sharp(t)|_{f^{-1}(V)} = \frac{\varphi(a)}{\varphi(s)}$, so $f^\sharp$ is indeed a well-defined morphism of sheaves.

We check that $(f, f^\sharp)$ is a morphism of locally ringed spaces, i.e. that $f^\sharp_\mathfrak{p} : \mathcal{O}_{Y,f(\mathfrak{p})} \to (f_*\mathcal{O}_X)_{f(\mathfrak{p})} \to \mathcal{O}_{X,\mathfrak{p}}$ is a local morphism of rings. By definition, this map is given by $[(U,t)] \mapsto [(f^{-1}(U), f^\sharp(t))]$, and hence by construction the diagram

$$\begin{tikzcd}
\mathcal{O}_{Y,f(\mathfrak{p})} \arrow[r, "f^\sharp_\mathfrak{p}"] \arrow[d, "\cong"] & \mathcal{O}_{X,\mathfrak{p}} \arrow[d, "\cong"] \\
A_{f(\mathfrak{p})} \arrow[r, "\varphi_\mathfrak{p}"] & B_\mathfrak{p}
\end{tikzcd}$$

commutes, where the vertical isomorphisms come from proposition 2.11. Since $\varphi_\mathfrak{p}$ is local, hence so is $f^\sharp_\mathfrak{p}$.

As was the case in classical algebraic geometry, we have

Proposition 2.18. *Let A, B be rings, and $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$. Then the construction above yields a bijection*

$$\operatorname{Hom}_{\texttt{Ring}}(A, B) \xrightarrow{\cong} \operatorname{Hom}_{\texttt{LRS}}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y)).$$

(In other words, the functor Spec *is fully faithful.)*

Proof. Let $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of locally ringed spaces. Then $f^\sharp_Y : \mathcal{O}_Y(Y) \cong A \to B \cong f_*\mathcal{O}_X(Y)$, we claim that this construction is inverse to the one constructed in remark 2.17. Given $\varphi : A \to B$, let $(f, f^\sharp)$ be the induced morphism of locally ringed spaces. Then $f^\sharp_Y = \varphi$ (under the canoncial identifications) by construction.

Conversely, let $(f, f^\sharp)$ be a morphism of locally ringed spaces, and let $(g, g^\sharp)$ be the morphism construed from $\varphi := f^\sharp_Y$ as above. Let $\mathfrak{p} \in X$. We have a commutative diagram

$$\begin{tikzcd}
B \cong \mathcal{O}_X(X) \arrow[d] & \mathcal{O}_Y(Y) \cong A \arrow[l, "f^\sharp_Y"'] \arrow[d] \\
B_\mathfrak{p} \cong \mathcal{O}_{X,\mathfrak{p}} & \mathcal{O}_{Y,f(\mathfrak{p})} \cong A_{f(\mathfrak{p})} \arrow[l, "f^\sharp_\mathfrak{p}"']
\end{tikzcd}$$

Since $f^\sharp_\mathfrak{p}$ is a local morphism, the preimage of $\mathfrak{p}B_\mathfrak{p}$ is the image of $f(\mathfrak{p})$ in $A_{f(\mathfrak{p})}$. By commutativity of the diagram, this equals $\varphi^{-1}(\mathfrak{p})$, i.e. $g = \varphi^{-1} = f$.

By the universal property of localizations, there is a unique map $A_{f(\mathfrak{p})} \to B_\mathfrak{p}$ making the above diagram commute. But both $f^\sharp_\mathfrak{p}$ and $\varphi_\mathfrak{p}$ do, so they are equal. By the construction in remark 2.17, this immediately implies $f^\sharp = g^\sharp$. $\square$

Definition 2.19. An *affine scheme* is a locally ringed space isomorphic to $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ for some ring A.

Remark 2.20. Immediately from the definition and proposition 2.18 it follows that the functor $\operatorname{Spec} : \texttt{Ring} \to \texttt{AffSch}$, $A \mapsto \operatorname{Spec} A$, $\varphi \mapsto (f, f^\sharp)$ as in remark 2.17 is an (anti-)equivalence of categories.

Definition 2.21. A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that for every $x \in X$ there exists a neighbourhood $x \in U \subseteq X$ such that $(U, \mathcal{O}_X|_U)$ is an affine scheme.

By abuse of notation, we will often write X instead of $(X, \mathcal{O}_X)$ and f instead of $(f, f^\#)$. Given a scheme X, we will write $\operatorname{Sp}(X)$ for the underlying topological space. Lecture 9 May 11, 2026

Lemma 2.22. *Let A be a ring and $X = \operatorname{Spec} A$. Let $g \in A$ and consider $D(g) \subseteq \operatorname{Spec} A$. Then $(D(g), \mathcal{O}_X|_{D(g)}) \cong \operatorname{Spec} A_g$ is again an affine scheme.*

Proof. Exercise. $\square$

Proposition 2.23. *Let X be a scheme and $U \subseteq X$ be open. Then $(U, \mathcal{O}_X|_U)$ is again a scheme.*

Proof. $(U, \mathcal{O}_X|_U)$ is clearly a locally ringed space. We only need to check that locally it is an affine scheme, so wlog $X = \operatorname{Spec} A$ is affine. Let $U = \bigcup_i D(g_i)$ be an open cover by standard opens, then each $(D(g_i), \mathcal{O}_X|_{D(g_i)}) = (D(g_i), \mathcal{O}_X|_U|_{D(g_i)})$ is an affine scheme by the previous lemma, so the result follows immediately. $\square$

Definition 2.24.
(i) Let X be a scheme and $U \subseteq X$ be open. Then $(U, \mathcal{O}_X|_U)$ is called an *open subscheme* of X. If moreover U is an affine scheme, then we say that U is an *affine open subset/subscheme* of X. Observe that by lemma 2.22, affine open subsets of X form a basis of the topology of X.
(ii) Consider the natural morphism $i_U : U \to X$ of schemes, where $(i_U^\sharp)_V : \mathcal{O}_X(V) \to i_*\mathcal{O}_X|_U(V) = \mathcal{O}_X(U \cap V)$ is the restriction. An *open immersion* is a morphism of schemes $Y \to X$ which induces an iosmorphism of Y with some open subscheme $U \subseteq X$, that is, it factors through i_U.
(iii) Let $f : X \to Y$ be a morphism of schemes and $U \subseteq X$ be open. Then we consider the restriction of f on U, $f|_U := f \circ i_U : U \to Y$.

Proposition 2.25 (Glueing of morphisms)**.** *Let X, Y be schemes and $X = \bigcup_{i \in I} U_i$ be an open cover. Then the map*

$$\operatorname{Hom}_{\texttt{Sch}}(X, Y) \to \prod_{i \in I} \operatorname{Hom}_{\texttt{Sch}}(U_i, Y), \qquad f \mapsto (f|_{U_i})_i$$

is injective, and has image consisting of those $(f_i)_i$ with $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all $i, j \in I$.

Proof. Let $(f, f^\sharp) : X \to Y$ be given. It is clear that f can be recovered uniquely from the $(f|_{U_i})_i$. Let $V \subseteq Y$ be open. Since $f_V^\sharp(s) \in \mathcal{O}_X(f^{-1}(V))$ is uniquely determined by its restrictions to $(f^{-1}(V) \cap U_i)_i$ by the sheaf condition, we can recover $f^\sharp$ as well.

It is clear that any family $(f_i)_i$ in the image satisfies the given condition. Conversely, suppose $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ holds for all $i, j \in I$. On the level of topological spaces the claim is clear: For $x \in X$ set $f(x) := f_i(x)$ for any i with $x \in U_i$. Now let $U \subseteq Y$ be open, $s \in \mathcal{O}_Y(U)$. The family of sections $t_i := (f_i)_U^\sharp(s) \in \mathcal{O}_{U_i}(U \cap U_i) = \mathcal{O}_X(U \cap U_i)$ glue to a unique section $f_U^\sharp(s)$, since $t_i|_{U_i \cap U_j} = (f_i)^\sharp_{U_i \cap U_j}(s) = (f_j)^\sharp_{U_i \cap U_j}(s) = t_j|_{U_i \cap U_j}$ by assumption. By uniqueness, this construction respects addition and multiplication, so $f_U^\sharp$ is a ring morphism. It is clear that $f^\sharp$ is a morphism of sheaves, and that $f|_{U_i} = f_i$, hence $f_x^\sharp$ is local for all $x \in X$, and $(f, f^\sharp)$ is a morphism of schemes as required. $\square$

3 Properties and some Constructions of Schemes

Definition 3.1. A scheme X is called *irreducible/connected* if its underlying topological space is.

Definition 3.2. A scheme X is called *reduced/integral* if for every open $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is reduced/an integral domain.

Remark 3.3. An affine scheme $X = \operatorname{Spec} A$ is irreducible if and only if the nilradical of A is prime. In particular, if A is an integral domain then X is irreducible. Furthermore, X is reduced/integral if and only if A is.

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Remark 3.4. Let X be a scheme. Then X is reduced if and only if $\mathcal{O}_{X,x}$ is reduced for every $x \in X$. If X is integral, then so are the stalks, but the converse fails in general.

Proposition 3.5. *Let X be a scheme. Then X is integral if and only if X is reduced and irreducible.*

Remark 3.6. In the affine case $X = \operatorname{Spec} A$, this follows from remark 3.3 and lemma 2.7.

Lemma 3.7. *Let X be a scheme and $f \in \mathcal{O}_X(X)$. Define $X_f := \{x \in X \mid f_x \notin \mathfrak{m}_x\}$, where $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is the maximal ideal of the local ring $\mathcal{O}_{X,x}$.*

(i) Let $U = \operatorname{Spec} B \subseteq X$ be open and $\widetilde{f} := f|_U \in \mathcal{O}_X(U) \cong B$. Then $X_f \cap U = D(\widetilde{f})$.
(ii) X_f is open in X.

Proof. (ii) follows from (i), since X can be covered by affine opens. Let $x \in U = \operatorname{Spec} B$. Then $f_x = \widetilde{f}_x$, so $x \in X_f \cap \operatorname{Spec} B$ if and only if $\widetilde{f}_x = \frac{\widetilde{f}}{1} \notin \mathfrak{m}_x = xB_x$, if and only if $\widetilde{f} \notin x$, i.e. $x \in D(\widetilde{f})$. $\square$

Proof of Proposition 3.5. First let X be integral. Then it is clear that X is reduced. Let $\emptyset = U_1 \cap U_2$ with U_1, U_2 open. By the sheaf property, $\mathcal{O}_X(U_1 \cup U_2) \cong \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$ is an isomorphism. Since $\mathcal{O}_X(U_1 \cup U_2)$ is integral by assumption, we must have $\mathcal{O}_X(U_1) = 0$, say, and hence $U_1 = \emptyset$.

Conversely, assume that X is reduced and irreducible. Let $U \subseteq X$ be open and $f, g \in \mathcal{O}_X(U)$ with $fg = 0$. Consider the open subsets $U_f, U_g \subseteq U$ from lemma 3.7. Then for $x \in U$, we have $f_x g_x = 0 \in \mathfrak{m}_x$, so $f_x \in \mathfrak{m}_x$ or $g_x \in \mathfrak{m}_x$. Hence $U_f \cap U_g = \emptyset$. Since X is irreducible, so is U, and wlog it follows that $U_f = \emptyset$. Showing that $f = 0$ can be done on an affine cover, so wlog $U = \operatorname{Spec} B$ is affine. Then lemma 3.7 says $D(f) = \emptyset$, and lemma 2.2 yields $f \in \operatorname{Nil} B = 0$. $\square$

Proposition 3.8. *Let X be an irreducible scheme. Then there exists a unique so-called* generic *point* $\xi \in X$ *such that* $\overline{\{\xi\}} = X$. *If X is integral, then $\mathcal{O}_{X,\xi}$ is a field, called its* function field.

Proof. If $X = \operatorname{Spec} A$ is affine, then $\overline{\{x\}} = V(x)$ by remark 2.5, and $\operatorname{Spec} A = V(x)$ is equivalent to $x = \operatorname{Nil} A$ by lemma 2.2. If A is integral, then $\xi = \operatorname{Nil} A = 0$, so $\mathcal{O}_{X,\xi} \cong A_{(0)} = \operatorname{Frac} A$ is a field.

For the general case, assume that ξ is a generic point. Then $\xi|_{\operatorname{Spec} A} = \operatorname{Nil} A$ for any affine open $\operatorname{Spec} A \subseteq X$, so ξ is uniquely determined by the sheaf condition. For existence, let ξ be the generic point of any affine open $U = \operatorname{Spec} A \subseteq X$, and let $V \subseteq X$ be open. Since X is irreducible, $U \cap V$ is a non-empty open in U, so contains ξ. Hence $\xi \in V$ for any $V \subseteq X$ open, i.e. $\overline{\{\xi\}} = X$.

Now let X be integral, and let $U = \operatorname{Spec} A \subseteq X$. Then by the previous paragraph, $0 \subseteq A$ is the generic point ξ of X, and $\mathcal{O}_{X,\xi} = \mathcal{O}_{U,0} = \operatorname{Frac} A$ is a field. $\square$

Remark 3.9. The proof shows: If X is integral with function field K, then $\operatorname{Frac} A \cong K$ for every affine open $\operatorname{Spec} A \subseteq X$. In particular, all affine opens in X have the same field of fractions.

Definition 3.10. A topological space X is *Noetherian* if X satisfies the descending chain condition for closed subsets, that is: Each chain $X \supseteq X_1 \supseteq X_2 \supseteq \dots$ with X_i closed stabilizes, i.e. there is $m \geq 1$ with $X_m = X_n$ for all $n \geq m$.

Proposition 3.11. *If A is a Noetherian ring, then $\operatorname{Spec} A$ is Noetherian.*

Remark 3.12. The converse is not true in general. For example, let k be an algebraically closed field and take $A = k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$.

Proof. Let $V(I_1) \supseteq V(I_2) \supseteq \dots$ be a descending chain of closed subsets in $\operatorname{Spec} A$. Since $V(J) = V(\sqrt{J})$ for any ideal J, wlog all the I_k are radical. Then by lemma 2.2, we get reverse inclusions $I_1 \subseteq I_2 \subseteq \dots$, which stabilize by assumption. ☐

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Definition 3.13. A topological space X is called *quasi-compact* if every open cover of X has a finite subcover. A scheme is quasi-compact if its underlying topological space is quasi-compact.

In some domains, quasi-compactness is usually just called compactness. In a more general setting, one sets compactness to mean quasi-compact and Hausdorff, which is why the difference rarely appears in fields related to Euclidean topologies. But the Zariski topology is almost never Hausdorff, so we use the term quasi-compactness to highlight this difference.

Lemma 3.14. *Let $X = \operatorname{Spec} A$ be an affine scheme. Then X is quasi-compact. Therefore, a scheme X is quasi-compact if and only if X can be covered by finitely many affine open subschemes.*

Proof. Let $X = \bigcup_{i \in I} U_i$ with U_i open. Let $X \setminus U_i = V(I_i)$, then $\emptyset = \bigcap_i V(I_i) = V(\sum_i I_i)$, which is equivalent to $1 \in \sum_i I_i$. But this means that 1 can be written as a *finite* linear combination of elements in the I_i, so those finitely many ideals suffice. ☐

Definition 3.15.
(i) A scheme X is *locally Noetherian* if there is an affine open cover $X = \bigcup_{i \in I} \operatorname{Spec} A_i$ such that all A_i are Noetherian rings.
(ii) X is *Noetherian* if there is a finite open cover $X = \bigcup_{i=1}^n \operatorname{Spec} A_i$ such that all A_i are Noetherian rings. Equivalently, X is Noetherian if it is locally Noetherian and quasi-compact.

Proposition 3.16. *If X is a Noetherian scheme, then its underlying space is Noetherian.*

Proof. $X = \bigcup_{i=1}^n \operatorname{Spec} A_i$ is a finite union of Noetherian topological spaces by proposition 3.11. Finite unions of Noetherian spaces are again Noetherian. ☐

Proposition 3.17.
(i) A topological space X is Noetherian if and only if every open $U \subseteq X$ is quasi-compact.
(ii) Let X be a Noetherian scheme and $U \subseteq X$ be open. Then U is also a Noetherian scheme.

Proof. Exercise. ☐

Proposition 3.18 (Nike's Lemma). *Let $\operatorname{Spec} A, \operatorname{Spec} B$ be two affine open subschemes of a scheme X. Then $\operatorname{Spec} A \cap \operatorname{Spec} B$ can be covered by open subsets which are simultaneously distinguished opens in $\operatorname{Spec} A$ and $\operatorname{Spec} B$.*

Proof. Let $x \in \operatorname{Spec} A \cap \operatorname{Spec} B$ and let $x \in D(h) \subseteq \operatorname{Spec} A \cap \operatorname{Spec} B$ be a principal neighbourhood in $\operatorname{Spec} A$. By the exercises, $D(h) \cong \operatorname{Spec} A_h \subseteq \operatorname{Spec} B$. Then find a distinguished open $x \in \operatorname{Spec} B_g \subseteq \operatorname{Spec} A_h$. These inclusions of affine schemes are induced by ring morphisms $\varphi : B \to A_h$ and $\psi : A_h \to B_g$ with $\psi \circ \varphi : B \to B_h$ being the canonical localization map. By sheet 4, exercise 1, $D(g) = D(\varphi(g))$ is distinguished in $\operatorname{Spec} A_h$, hence also in $\operatorname{Spec} A$, since if $\varphi(g) = \frac{a}{h^m}$, then $D(\varphi(g)) = D(h) \cap D(a) = D(ha)$ □

Lemma 3.19 (Affine Communication Lemma)**.** *Let X be a scheme and $\mathcal{P}$ be a property of open affine subschemes of X, such that*

(i) if $\operatorname{Spec} A \hookrightarrow X$ has $\mathcal{P}$, then so does every $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A \hookrightarrow X$, and
(ii) if $(f_1, \dots, f_n) = A$ and $\operatorname{Spec} A_{f_i} \hookrightarrow \operatorname{Spec} A \hookrightarrow X$ have $\mathcal{P}$, then so does $\operatorname{Spec} A$.

Assume $X = \bigcup_{i \in I} \operatorname{Spec} A_i$ with all $\operatorname{Spec} A_i$ satisfying $\mathcal{P}$. Then every affine open subset of X satisfies $\mathcal{P}$.

Proof. Let $\operatorname{Spec} A \subseteq X$ be an affine open. By Proposition 3.18 we can write $\operatorname{Spec} A = \bigcup_{j \in J} U_j$ such that U_j is distinguished both in $\operatorname{Spec} A$ and in some $\operatorname{Spec} A_i$. By (i) all U_j satisfy $\mathcal{P}$. Since $\operatorname{Spec} A$ is quasi-compact by lemma 3.14, we can find a finite subcover, and the result follows from (ii). □

Proposition 3.20. *Let X be a scheme. Then X is locally Noetherian if and only if for every affine open $\operatorname{Spec} A \subseteq X$, A is a Noetherian ring.*

Corollary 3.21. $\operatorname{Spec} A$ *is a Noetherian scheme if and only if A is Noetherian.*

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Proof (of Proposition 3.20). We have to check conditions (i),(ii) of Lemma 3.19 for the property $\mathcal{P}(\operatorname{Spec} A) = A$ is Noetherian. Since every localization of a Noetherian ring is again Noetherian, (i) is clear. So let $(f_1, \dots, f_n) = A$ and assume that all A_{f_i} are Noetherian. Write $\varphi_i : A \to A_{f_i} =: S_i^{-1} A$ for the canonical localization maps. Let $I \subseteq A$ be an ideal, then $I = \bigcap_{i=1}^n \varphi_i^{-1}(S_i^{-1} I)$: Indeed, "$\subseteq$" is clear, so assume $a \in \bigcap_{i=1}^n \varphi_i^{-1}(S_i^{-1} I) = \bigcap_{i=1}^n I^{S_i}$. It follows that $f_i^{k_i} a \in I$ for suitable $k_1, \dots, k_n \in \mathbb{N}$. Since $1 = (f_1, \dots, f_n) = (f_1^{k_1}, \dots, f_n^{k_n})$, we get $a \in (f_1^{k_1} a, \dots, f_n^{k_n} a) \subseteq I$.

Let $I_1 \subseteq I_2 \subseteq \dots$ be an ascending chain of ideals in A. For every $i = 1, \dots, n$, the localized chain $S_i^{-1} I_1 \subseteq S_i^{-1} I_2 \subseteq \dots$ stabilizes by assumption. Once these finally many chains have all stabilized, by the above formula so does the original chain. □

Let A be a ring. A scheme over A (or A-scheme) is a scheme X with a morphism $X \to \operatorname{Spec} A$. An affine open subset $\operatorname{Spec} B \subseteq X$ thus induces a ring map $A \to B$, so B can be considered an A-algebra.

Definition 3.22. Let X be a scheme over A.

(i) X is *locally of finite type* over A if $X = \bigcup_i \operatorname{Spec} B_i$, such that B_i is a finitely generated A-algebra.
(ii) X is of *finite type* over A if X is locally of finite type and quasi-compact, equivalently if $X = \bigcup_{i=1}^n \operatorname{Spec} B_i$ with all B_i finitely generated A-algebras.

Proposition 3.23. *A scheme X is locally of finite type over A if and only if for every affine open $\operatorname{Spec} B \subseteq X$, B is a finitely generated A-algebra.*

Proof. Exercise (use lemma 3.19 again). □

Corollary 3.24. $\operatorname{Spec} B$ *is of finite type over A if and only if B is a finitely generated A-algebra.*

Definition 3.25. Let K be a field. An affine scheme X which is reduced and of finite type over K is called an *affine variety* over K.

In the literature, for an affine variety over K sometimes it is required that K is algebraically closed. A general variety over K can be defined as a reduced scheme of finite type over K. A more common definition also requires the scheme to be separated, more on this later.

First we want to have a notion of projective spaces in our language of schemes. Let $B = \bigoplus_{i\geq 0} B_i$ be a graded ring. Elements of some B_i are called homogenoues elements of degree i. Set $B_+ := \bigoplus B_i$. An ideal $I \subseteq B$ is called homogeneous if it is generated by homogeneous elements, equivalently if $b = b_0 + b_1 + \ldots + b_n$ is the decomposition into a sum of graded components, then $b \in I$ if and only if $b_i \in I$ for all i. A homogeneous ideal $\mathfrak{p} \subseteq B$ is prime if and only if $bc \in \mathfrak{p} \Rightarrow b \in \mathfrak{p}$ or $c \in \mathfrak{p}$ for all homogeneous $b, c \in B$.

Let $\operatorname{Proj} B$ be the set of all homogeneous prime ideals $\mathfrak{p} \subseteq B$ with $B_+ \not\subseteq \mathfrak{p}$. For a homogeneous ideal $I \subseteq B$, set $V_+(I) := \{\mathfrak{p} \in \operatorname{Proj} B \mid I \subseteq \mathfrak{p}\}$. As in the affine case, we have

Lemma 3.26. *If $I, J, (I_\alpha)_{\alpha\in A}$ are homogeneous ideals, then $V_+(IJ) = V_+(I) \cup V_+(J)$ and $V_+(\sum_\alpha I_\alpha) = \bigcap_\alpha V_+(I_\alpha)$.* □

By lemma 3.26 we can define the *Zariski* topology on $\operatorname{Proj} B$ by declaring closed subsets to be the subsets of the form $V_+(I)$ for homogeneous ideals $I \subseteq B$.

Remark 3.27. Let $I = (\{f_j\}_j)$ be a homogeneous ideal, with a homogeneous generating set $\{f_j\}_j$. Then $V_+(I) = \emptyset$ if and only if $\operatorname{Proj} B = \bigcup_j D_+(f_j)$ if and only if $B_+ \subseteq \sqrt{I}$ (Exercise).

Definition 3.28. For any homogeneous element $f \in B_+$ we define $D_+(f) = \{\mathfrak{p} \in \operatorname{Proj} B \mid f \notin \mathfrak{p}\} = \operatorname{Proj} B \setminus V_+((f))$. As in the affine case, we have $D_+(f) \cap D_+(g) = D_+(fg)$ and the $\{D_+(f)\}_f$ form a basis of the topology on $\operatorname{Proj} B$.

Remark 3.29. Let $\mathfrak{p} \in \operatorname{Proj} B$. Let $T_\mathfrak{p} \subseteq B$ be the set of homogeneous elements not in $\mathfrak{p}$. Then $T_\mathfrak{p}^{-1}$ is a ($\mathbb{Z}$-)graded ring with $\deg(\frac{a}{s}) := \deg a - \deg s$. Set $B_{(\mathfrak{p})} \subseteq T_\mathfrak{p}^{-1}B$ to be the subring of elements of degree 0, i.e. $B_{(\mathfrak{p})} = \{\frac{a}{s} \mid a, s \text{ homogeneous}, s \notin \mathfrak{p} \text{ and } \deg(a) = \deg(s)\}$. $B_{(\mathfrak{p})}$ is a local ring with maximal ideal $\mathfrak{n}_{(\mathfrak{p})} := B_{(\mathfrak{p})} \cap \mathfrak{p}T_\mathfrak{p}^{-1}B$, for if $\frac{a}{b} \in B_{(\mathfrak{p})} \setminus \mathfrak{n}_{(\mathfrak{p})}$, then $a \notin \mathfrak{p}$ and $\frac{b}{a}$ still has degree 0, so $\frac{b}{a} \in B_{(\mathfrak{p})}$ and $\frac{a}{b}$ is a unit.

Definition 3.30. Let $X = \operatorname{Proj} B$. We define the *structure sheaf* of X as

$$\mathcal{O}_X(U) := \left\{ s : U \to \coprod_{\mathfrak{p}\in U} B_{(\mathfrak{p})} \mid s(\mathfrak{p}) \in B_{(\mathfrak{p})} \text{ for all } \mathfrak{p} \in U, \text{ and } (*) \right\},$$

where $(*)$ is the condition that for any $\mathfrak{q} \in U$ there exists an open neighbourhood $\mathfrak{q} \in V \subseteq U$ and homogeneous elements $f, g \in B$ of the same degree such that $g \notin \mathfrak{p}$ and $s(\mathfrak{p}) = \frac{f}{g} \in B_{(\mathfrak{p})}$ for all $\mathfrak{p} \in V$.

Clearly $(X, \mathcal{O}_X)$ is a ringed space. By the same proof as in proposition 2.11, $\mathcal{O}_{X,\mathfrak{p}} \cong B_{(\mathfrak{p})}$, so by remark 3.29 $(X, \mathcal{O}_X)$ is a locally ringed space. Lecture 13 Jun 01, 2026

Theorem 3.31. *$(X, \mathcal{O}_X)$ is a scheme. More precisely, for every $f \in B_+$ we have an isomorphism $(D_+(f), \mathcal{O}_{D_+(f)}) \cong (\operatorname{Spec} B_{(f)}, \mathcal{O}_{\operatorname{Spec} B_{(f)}})$, where $B_{(f)} \subseteq B_f$ is the subring of elements of degree 0.*

Example 3.32. Let $B = K[x_0, \ldots, x_n]$ with the usual grading, $X = \mathbb{P}^n_K := \operatorname{Proj} B$ and $f = x_i$. Then $B_{(x_i)} \cong k[x_0, \ldots, \widehat{x}_i, \ldots, x_n]$ via inhomogenization, so theorem 3.31 says that $\mathbb{P}^n_K$ is covered by copies of $\mathbb{A}^n_K \cong D_+(x_i)$, in accordance with the classical situation.

Proof of Theorem 3.31. Note that $\operatorname{Proj} B$ is a subspace of $\operatorname{Spec} B$, since if $f = f_0 + \ldots + f_r \in B$, then $D(f) \cap \operatorname{Proj} B = \bigcup_i D_+(f_i)$. In particular, if f is homogeneous, then $D(f) \cap \operatorname{Proj} B = D_+(f)$, so we get a continuous map $\theta : D_+(f) \hookrightarrow D(f) \cong \operatorname{Spec} B_f \to \operatorname{Spec} B_{(f)}$, $\mathfrak{p} \mapsto \mathfrak{p}B_f \cap B_{(f)}$. We claim that θ is a homeomorphism. θ is open, since if $D_+(g) \subseteq D_+(f)$, then $\theta(D_+(g)) = D(h)$ with $h = \frac{g^{\deg f}}{f^{\deg g}}$. (Exercise.) So it remains to show that θ is bijective.

Let $d := \deg f$. For $\mathfrak{q} \in \operatorname{Spec} B_{(f)}$ set $\psi(\mathfrak{q}) := \bigoplus_{n \geq 0} \psi(\mathfrak{q})_n$, where $\psi(\mathfrak{q})_n := \{b \in B_n \mid \frac{b^d}{f^n} \in \mathfrak{q}\}$. $\psi(\mathfrak{q})$ is a graded ideal in B: Let $a \in B_m$ and $b \in \psi(\mathfrak{q})_n$, then $\frac{(ab)^d}{f^{m+n}} = \frac{a^d}{f^m}\frac{b^d}{f^n} \in \mathfrak{q}$, so $ab \in \psi(\mathfrak{q})_{m+n}$. Also if $a, b \in \psi(\mathfrak{q})_n$, then $(\frac{(a+b)^d}{f^n})^2 = \frac{1}{f^{2d}} \sum_i \binom{2d}{i} a^i b^{2d-i} \in \mathfrak{q}$, and since $\mathfrak{q}$ is prime, $\frac{(a+b)^d}{f^n} \in \mathfrak{q}$, i.e. $a + b \in \psi(\mathfrak{q})_n$. Furthermore, $f \notin \psi(\mathfrak{q})$, since otherwise by construction $\frac{f^d}{f^d} = 1 \in \mathfrak{q}$. To see that $\psi(\mathfrak{q})$ is prime, let $a \in B_m$, $b \in B_n$ with $ab \in \psi(\mathfrak{q})_{n+m}$. That is, $\frac{a^d b^d}{f^m f^n} \in \mathfrak{q}$, so $\frac{a^d}{f^m} \in \mathfrak{q}$ or $\frac{b^d}{f^n} \in \mathfrak{q}$, which exactly means $a \in \psi(\mathfrak{q})_m$ or $b \in \psi(\mathfrak{q})_n$. Thus we have successfully constructed a map $\psi : \operatorname{Spec} B_{(f)} \to D_+(f)$ in the inverse direction, and one checks that it is indeed inverse to θ (Exercise).

It remains to construct an isomorphism $\theta^\sharp : \mathcal{O}_{\operatorname{Spec} B_{(f)}} \to \theta_* \mathcal{O}|_{D_+(f)}$. For this, note that for $\mathfrak{p} \in D_+(f)$, we have natural isomorphisms $\varphi_\mathfrak{p} : (B_{(f)})_{\theta(\mathfrak{p})} \to B_{(\mathfrak{p})}$, $\frac{a/f^n}{b/f^m} \mapsto \frac{af^m}{bf^n}$ from universal properties of localization. Indeed, $\frac{b}{f^n} \notin \theta(\mathfrak{p})$ if and only if $b \notin \mathfrak{p}$ for $b \in B_{dn}$, injectivity of $\varphi_\mathfrak{p}$ is clear, and for $\frac{a}{b} \in B_{(f)}$ one has $\varphi_\mathfrak{p}(\frac{ab^{d-1}/f^{\deg a}}{b^d/f^{\deg b}}) = \frac{a}{b}$. Hence we can identify the sections of the two sheaves, i.e. define $\theta^\sharp_{\theta(U)} : \mathcal{O}_{\operatorname{Spec} B_{(f)}}(\theta(U)) \to \mathcal{O}_{\operatorname{Proj} B}(U)$, $s \mapsto (\mathfrak{p} \mapsto \varphi_\mathfrak{p}(s(\theta(\mathfrak{p}))))$, which is clearly an isomorphism for any $U \subseteq D_+(f)$ open. □ Lecture 14 Jun 8, 2026

Example 3.33. (i) Let B be a graded R-algebra, i.e. $\varphi : R \to B$ with $\varphi(R) \subseteq B_0$. Then $\operatorname{Proj} B$ is a scheme over R, since all localizations $B_{(\mathfrak{p})}$ are R-algebras.

(ii) For any ring R set $\mathbb{P}^n_R := \operatorname{Proj} R[x_0, \ldots, x_n]$ with the usual grading (cf. example 3.32), which is a scheme over R by (i). Clearly, $\mathbb{P}^n_R$ is a scheme of finite type over R.

Definition 3.34. (i) Let $s_X : X \to \operatorname{Spec} R, s_Y : Y \to \operatorname{Spec} R$ be schemes over a ring R. A *morphism of R-schemes* is a morphism of schemes $f : X \to Y$ compatible with the structure maps, i.e. $s_X = s_Y \circ f$. In this way, R-schemes form a category, with terminal object $\mathrm{id} : \operatorname{Spec} R \to \operatorname{Spec} R$.

(ii) Let X be a scheme over R. The *R-points* of X are $X(R) := \operatorname{Hom}_{R-\mathtt{Sch}}(\operatorname{Spec} R, X)$. A morphism $f : X \to Y$ of R-schemes induces a map $X(R) \to Y(R)$ of sets.

(iii) Let X be a scheme and $x \in X$. The *residue field* of x is $k(x) := \mathcal{O}_{X,x}/\mathfrak{m}_x$.

If k is a field, and X is a scheme over k, then $X(k) = \{x \in X \mid k(x) = k\}$ (Exercise). If $X = \operatorname{Spec} k[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$, there is a bijection $X(k) \cong \{\alpha \in k^n \mid f_i(\alpha) = 0$ for all $i = 1, \ldots, m\}$ (Exercise), i.e. we can "recover" the classical affine variety from X. The same holds in the projective case:

Remark 3.35. (i) Let k be a field. $\mathbb{P}^n_k(k) \cong (k^{n+1} \setminus \{0\})/(x \sim \lambda x)$ can be identified with the underlying set of classical projective space (Exercise). As above, one can consider more generally $X = \operatorname{Proj} k[X_0, \ldots, X_n]/(f_1, \ldots, f_m)$ (with f_i graded), then $X(k) = \{x = [x_0 : \ldots : x_n] \mid f_1(x) = \ldots = f_m(x) = 0\}$ under the same identification.

(ii) If B is integral, then $\operatorname{Proj} B$ is an integral scheme. In particular, if $I \subseteq k[X_0, \ldots, X_n]$ is prime, then $\operatorname{Proj} k[X_0, \ldots, X_n]/I$ is an integral noetherian scheme of finite type over k.

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Proposition 3.36. *Let $\varphi : C \to B$ be a homomorphism of graded rings (i.e. there is $r \geq 1$ such that $\varphi(C_d) \subseteq B_{rd}$ for all d). Consider the ideal $J = \langle \varphi(C_+) \rangle$. Then φ induces a morphism* *of schemes* $f : \operatorname{Proj} B \setminus V_+(J) \to \operatorname{Proj} C$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. *Moreover, for* $h \in C_+$ *we have* $f^{-1}(D_+(h)) = D_+(\varphi(h))$ *and*

$$\begin{array}{ccc} D_+(\varphi(h)) & \xrightarrow{f} & D_+(h) \\ \downarrow{\cong} & & \downarrow{\cong} \\ \operatorname{Spec} B_{(\varphi(h))} & \xrightarrow{\operatorname{Spec}\overline{\varphi}} & \operatorname{Spec} C_{(h)} \end{array}$$

commutes, where the vertical arrows are the isomorphisms from proposition 3.31 and $\overline{\varphi} : C_{(h)} \to B_{(\varphi(h))}$ *is the map induced from* φ *by localization.*

Proof. Exercise (similar to the affine case). □

Example 3.37. (i) Let $\varphi : K[X_0, X_1] \to K[X_0, X_1, X_2]$ be the inclusion. In the notation of proposition 3.35, $J = (X_0, X_1) \subseteq K[X_0, X_1, X_2]$ (corresponding to the K-rational point $(0 : 0 : 1)$), so we get a morphism $\mathbb{P}^3_K \setminus \{J\} \to \mathbb{P}^2_K$. On K-rational points it is given as $[a : b : c] \mapsto [a : b]$.

(ii) Let $\varphi : K[X_0, X_1, X_2] \to K[X_0, X_1]$ be given by $X_2 \mapsto 0$. Then $J = B_+$, so we obtain a map $\mathbb{P}^2_K \to \mathbb{P}^3_K$, which on K-points is $[a : b] \mapsto [a : b : 0]$.

4 Closed Subschemes and Fibred Products of Schemes

Example 4.1. Let A be a ring, $I \subseteq A$ be an ideal. Then $A \to A/I$ induces a morphism of schemes $f : \operatorname{Spec} A/I \to \operatorname{Spec} A$, with image $V(I)$, which is closed.

Remark 4.2. Closed subsets of $\operatorname{Spec} A$ are of the form $V(I)$. By example 4.1, we can define a scheme structure on $V(I)$, namely $(V(I), f_*\mathcal{O}_{\operatorname{Spec} A/I}) \cong \operatorname{Spec} A/I$. However, for another ideal $J \neq I$ we may have $V(I) = V(J)$ and we get another scheme structure on $V(I)$.

Remark 4.3. A closed immersion (or closed embedding) should intuitively be a morphism $f : Z \to X$ of schemes such that f induces a homeomorphism $f : Z \to f(Z) \subseteq X$ with $f(Z)$ closed in X, and if $X = \bigcup_i U_i$ is an open affine cover, $U_i = \operatorname{Spec} A_i$, then $f^{-1}(U_i) \cong \operatorname{Spec} A_i/I$ for some ideal $I \subseteq A_i$, and $f|_{f^{-1}(U_i)}$ looks like $\operatorname{Spec} A_i/I \to \operatorname{Spec} A$. For practical purposes, we will give an a priori different definition, which will turn out to be equivalent:

Definition 4.4. A *closed immersion* is a morphism $f : Z \to X$ of schemes such that f induces a homeomorphism $f : Z \to f(Z)$ with $f(Z) \subseteq X$ closed, and $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Z$ is surjective.

Example 4.5. Let $I \subseteq A$ be an ideal. The morphism $f : \operatorname{Spec} A/I \to \operatorname{Spec} A$ from example 4.1 is a closed immersion.

Definition 4.6. A *closed subscheme* of a scheme X is an equivalence class of closed immersions $f : Z \to X$, where f is equivalent to $f' : Z' \to X$ if and only if there is an isomorphism $g : Z \to Z'$ with $f = f' \circ g$.

Remark 4.7. Equivalently, a closed subscheme of X is given by a closed subset $Z \subseteq X$ with a scheme structure $(Z, \mathcal{O}_Z)$ such that there is a closed immersion $Z \to X$ which is the inclusion on the level of topological spaces.

Remark 4.8. In the affine case, $\operatorname{Spec} A/I \to V(I) \subseteq \operatorname{Spec} A$ is a closed subscheme.

Proposition 4.9. *Let $X = \operatorname{Spec} A$ be an affine scheme and $f : Z \to X$ be a closed immersion. Then there exists a unique ideal $I \subseteq A$ such that $f : Z \to X$ is equivalent to $\operatorname{Spec} A/I \to X$ in the sense of definition 4.6.*

Corollary 4.10. *$f : Z \to X$ is a closed immersion if and only if for every affine open $\operatorname{Spec} A \cong U \subseteq X$, the restriction $f|_{f^{-1}(U)} : f^{-1}(U) \to \operatorname{Spec} A$ is equivalent to $\operatorname{Spec} A/I \to \operatorname{Spec} A$ for some ideal $I \subseteq A$, as closed immersions to $\operatorname{Spec} A$.*

Remark 4.11. Let X be a scheme and $f \in \mathcal{O}_X(X)$. Recall from lemma 3.7 the open set $X_f = \{x \in X \mid f_x \notin \mathfrak{m}_x\} = \{x \in X \mid f_x \neq 0 \in k(x)\}$. Let $\varphi : X \to \operatorname{Spec} A$ be a morphism of schemes and $g \in A$. Then $\varphi^{-1}(D(g)) = X_{\varphi^\sharp_{\operatorname{Spec} A}(g)}$. Indeed, this can be checked locally and is clear from the construction if X is affine (see sheet 4, exercise 1).

Lemma 4.12. *Let X be a scheme and $X = \bigcup_{i=1}^n U_i$ be a finite affine open cover such that $U_i \cap U_j$ admits a finite affine open cover for every i, j. (For example if X is affine or Noetherian.) Then for $f \in \mathcal{O}_X(X)$, the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(X_f)$ induces an isomorphism $\mathcal{O}_X(X)_f \to \mathcal{O}_X(X_f)$.*

Proof. Exercise. ☐

Lemma 4.13 (Affineness criterion)**.** *Let X be a scheme. Assume that there exist $f_1, \dots, f_r \in \mathcal{O}_X(X)$ such that $(f_1, \dots, f_r) = \mathcal{O}_X(X)$ and X_{f_i} are affine for all $i = 1, \dots, r$. Then X is affine.*

Proof. Note that $X = \bigcup_{i=1}^r X_{f_i}$ since for every $x \in X$, the formula $(f_1, \dots, f_r) = 1$ also holds in $k(x)$, so not all of them can vanish at x.

Since X_{f_i} is affine, so is $X_{f_i} \cap X_{f_j}$ by lemma 3.7. Thus the requirements of lemma 4.12 are satisfied for this cover, and we get isomorphisms $\mathcal{O}_X(X)_{f_i} \to \mathcal{O}_X(X_{f_i})$. Write $A := \mathcal{O}_X(X)$ and $X_{f_i} = \operatorname{Spec} A_i$. Then the lemma identifies the restrictions $A \to A_i$ with the localizations $A \to A_{f_i}$. By sheet 4, exercise 4 we get a morphism $g : X \to \operatorname{Spec} A$ corresponding to id_A. By the above, the compositions $\operatorname{Spec} A_{f_i} \cong X_{f_i} \to X \to \operatorname{Spec} A$ are induced by the canonical localization maps $A \to A_{f_i}$. Lecture 16 Jun 15, 2026

By remark 4.11 we get $g^{-1}(D(f_i)) = X_{f_i}$, so $g|_{X_{f_i}} : X_{f_i} \to D(f_i)$ is an isomorphism. It follows from $\bigcup_{i=1}^r D(f_i) = \operatorname{Spec} A$ and the lemma below that g is an isomorphism. $\square$

Lemma 4.14. *Let $f : X \to Y$ be a morphism of schemes and $Y = \bigcup_i V_i$ an open cover. Then f is an isomorphism if and only if $f|_{f^{-1}(V_i)} : f^{-1}(V_i) \to V_i$ are isomorphisms for all i.*

Proof. The claim is clear on the level of topological spaces. For the maps of sheaves, it follows from the fact that isomorphisms can be checked on stalks (by proposition 1.11). $\square$

Proof of Proposition 4.9. Cover Z by affine opens $\operatorname{Spec} B_i \cong f^{-1}(U_i)$ for some $U_i \subseteq \operatorname{Spec} A$. Let $D(g_j) \subseteq U_i$. Write $\varphi = f^\sharp_{\operatorname{Spec} A} : A \to \mathcal{O}_Z(Z)$, then $f^{-1}(D(g_j)) = X_{\varphi(g_j)} \subseteq \operatorname{Spec} B_i$ is affine by remark 4.11 and lemma 3.7. Since $Z \simeq f(Z)$ is quasi-compact (as closed subset of a quasi-compact space), finitely many such $X_{\varphi(g_1)}, \dots, X_{\varphi(g_n)}$ cover Z. Now $\operatorname{Spec} A$ is covered by $D(g_1), \dots, D(g_n)$ and some $D(g'_k)$ that cover $X \setminus f(Z)$. Then $(g_i, g'_k)_{i,k} = 1$ in A and hence $(\varphi(g_i), \varphi(g'_k))_{i,k} = 1$ in $\mathcal{O}_Z(Z)$. Moreover, $f^{-1}(D(g'_k)) = \emptyset$, so $\varphi(g'_k)$ is nilpotent (since the restriction to any affine open is). Therefore $(\varphi(g_i))_i = 1$ and we can apply lemma 4.13 to deduce that Z is affine.

Write $Z = \operatorname{Spec} B$, it remains to show that the corresponding map $\psi : A \to B$ is surjective. Since ψ factors through $A/\ker\psi$, f factors through $\theta : \operatorname{Spec} B \to \operatorname{Spec} A/I$.

$$\begin{tikzcd} B & & A \arrow[ll, "\psi"'] \arrow[dl] \\ & A/J \arrow[ul] & \end{tikzcd} \qquad \begin{tikzcd} \operatorname{Spec} B = Z \arrow[rr, hook, "f"] \arrow[dr, "\theta"] & & \operatorname{Spec} A \\ & \operatorname{Spec} A/\ker\psi \arrow[ur, hook] & \end{tikzcd}$$

It follows that θ is injective (on topological spaces) and closed. By the following lemma, $\operatorname{im}\theta$ is dense, and hence θ is a homeomorphism. The lemma also implies that $\theta^\sharp$ is injective, and surjectivity follows from the diagram above and the definition of closed immersions. $\square$

Lemma 4.15. *Let $\varphi : A \to B$ be a ring homomorphism and $(f, f^\sharp) : \operatorname{Spec} B \to \operatorname{Spec} A$ the corresponding morphism of schemes. Then φ is injective if and only if $f^\sharp$ is injective. In this case, f is dominant, i.e. $\operatorname{im} f$ is dense in $\operatorname{Spec} A$.*

Proof. Exercise. $\square$

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Remark 4.16. Let X be a scheme and $Z \subseteq X$ be a closed subset. We want to introduce a scheme structure $\mathcal{O}_Z$ on Z so that $Z \to X$ becomes a closed immersion. In the affine case, let $Z = V(I)$. Then we may take $\operatorname{Spec} A/J$ for any J with $Z = V(J)$, so there may be different closed scheme structures. However, only $J = \sqrt{I}$ yields a reduced subscheme.

The same is true in the general case: $Z \subseteq X$ can be endowed with a unique reduced closed subscheme structure: Let $X = \bigcup_i V_i$ with $V_i \cong \operatorname{Spec} A_i$. Then $Y_i := Z \cap V_i$ is closed in V_i, so $Y_i = V(J_i)$ for a unique radical ideal $J_i \subseteq A_i$. Giving Y_i the reduced subscheme structure $\operatorname{Spec} A_i/J_i \to \operatorname{Spec} A_i$ as above, one can check that these glue to a scheme structure on Z (cf. 4.21), which is still reduced by remark 3.4.

Example 4.17. Let K be a field, $X = \mathbb{A}^2_K$, $Z = \{(0,0)\} \subseteq X$. Then $J_1 = (X, Y)$ and $J_2 = (X^2, Y)$ give two different closed subschemes $Z_i := \operatorname{Spec} K[X_1, X_2]/J_i \to X$, with Z_1 being the reduced one.

Remark 4.18 (Intersection of closed subschemes)**.** In the affine case $X = \operatorname{Spec} A$, let $Z_1 = \operatorname{Spec} A/J_1$, $Z_2 = \operatorname{Spec} A/J_2$ be two closed subschemes. Since $V(J_1) \cap V(J_2) = V(J_1 + J_2)$, we define $Z_1 \cap Z_2 = \operatorname{Spec} A/(J_1 + J_2)$. For example, the intersection of the parabola $V(Y - X^2)$ and the X-axis $V(Y) \subseteq \mathbb{A}^2_K$ is $\operatorname{Spec} K[X,Y]/(Y, Y - X^2) = \operatorname{Spec} K[X,Y]/(Y, X^2) = \operatorname{Spec} K[X]/(X^2)$, so we can "see" that the intersection has multiplicity 2.

In general, if $Z_1, Z_2 \subseteq X$ are two closed subschemes, then $Z_1 \cap Z_2$ can be defined in a way that for every affine open $\operatorname{Spec} A \subseteq X$, the scheme structure agrees with the construction above.

Remark 4.19. Let $B = A[X_0, \dots, X_n]$ and $X = \mathbb{P}^n_A = \operatorname{Proj} B$. Then every closed subscheme of X is of the form $\operatorname{Proj} A[X_0, \dots, X_n]/I$ for some homogeneous ideal $B_+ \not\subseteq I \subseteq B$.

Proposition 4.20 (Glueing sheaves). *Let X be a topological space, $X = \bigcup_i U_i$ an open cover. Assume the following* glueing data *is given:*

(i) A sheaf $\mathcal{F}_i$ on every U_i,
(ii) For every i, j an isomorphism $\varphi_{ij} : \mathcal{F}_i|_{U_i \cap U_j} \to \mathcal{F}_j|_{U_i \cap U_j}$ such that $\varphi_{ii} = \mathrm{id}_{\mathcal{F}_i}$ and $\varphi_{ik}|_{U_i \cap U_k \cap U_j} = \varphi_{jk}|_{U_j \cap U_k \cap U_i} \circ \varphi_{ij}|_{U_i \cap U_j \cap U_k}$ for all i, j, k.

Then there exists a unique sheaf $\mathcal{F}$ on X together with isomorphisms $\psi_i : \mathcal{F}|_{U_i} \to \mathcal{F}_i$ such that $\psi_j|_{U_i \cap U_j} = \varphi_{ij} \circ \psi_i|_{U_i \cap U_j}$. We say that $\mathcal{F}$ is obtained by glueing the sheaves $\mathcal{F}_i$ along the φ_{ij}.

Proof. Set $\mathcal{F}(V) := \{(s_i)_i \in \mathcal{F}_i(U_i \cap V) \mid \varphi_{ij}(s_i|_{V \cap U_i \cap U_j}) = s_j|_{V \cap U_i \cap U_j}\}$. One checks that this satisfies all requirements (Exercise). ☐

Remark 4.21. If $(U_i, \mathcal{F}_i)$ are schemes and the φ_{ij} are morphisms of schemes, then $(X, \mathcal{F})$ is a scheme as well.

Proposition 4.22. *Let $(X_i)_{i \in I}$ be schemes. Given open subsets $X_{ij} \subseteq X_i$ and isomorphisms $\varphi_{ij} : X_{ij} \to X_{ji}$ of schemes for all i, j such that $\varphi_{ij}(X_{ij} \cap X_{ik}) = X_{ji} \cap X_{jk}$ and $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $X_{ij} \cap X_{ik}$ for all i, j, k.* Lecture 18 Jun 22, 2026

Then there exists a unique scheme X with an open cover $X = \bigcup U_i$ and isomorphisms $\psi_i : U_i \to X_i$ of schemes such that $\psi_j = \varphi_{ij} \circ \psi_i$ on $U_i \cap U_j = \psi_i^{-1}(X_{ij})$.

Proof. Glue the topological spaces X_i along the φ_{ij} to a space $X = \coprod X_i / \langle x_i \sim \varphi_{ij}(x_i) \rangle$. Upon identifying the X_i with their homeomorphic image U_i in X, and transporting the sheaf structures, we are exactly in the situation of proposition 4.20. ☐

Remark 4.23. By proposition 2.25, a morphism $f : X \to Y$ with X as above is uniquely determined by morphisms $f_i : X_i \to Y$ such that $f_i = f_j \circ \varphi_{ij}$ on $X_{ij} \subseteq X_i$.

Example 4.24. Let R be a ring. By example 3.32, we have $\mathbb{P}^n_R = \bigcup D_+(x_i)$ with $D_+(x_i) = \operatorname{Spec} K[x_0, \dots, \widehat{x}_i, \dots, x_n] \cong \mathbb{A}^n_K$. Hence $\mathbb{P}^n_R$ could alternatively be obtained by glueing these $\mathbb{A}^n_K$ along the morphisms induced by the ring maps

$$\varphi^\sharp_{ij}(D_+(x_i x_j)) : R[x_0, \dots, x_j = 1, \dots, x_n]_{x_i} \to R[x_0, \dots, x_i = 1, \dots, x_n]_{x_j}, \quad x_k \mapsto \frac{x_k}{x_j}.$$

Definition 4.25. Let $\mathcal{C}$ be any category and $f : X \to S$, $g : Y \to S$ be two morphisms in $\mathcal{C}$. The fibred product $X \times_S Y$ is an object in $\mathcal{C}$ together with projection morphisms $p_1 : X \times_S Y \to X$ and $p_2 : X \times_S Y \to Y$ such that $f \circ p_1 = g \circ p_2$ and for every object Z with morphisms $\alpha : Z \to X$,

$\beta : Z \to Y$ such that $f \circ \alpha = g \circ \beta$, there exists a unique morphism $\theta : Z \to X \times_S Y$ such that $p_1 \circ \theta = \alpha$ and $p_2 \circ \theta = \beta$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "\exists!\theta"] \arrow[drr, bend left, "\beta"] \arrow[ddr, bend right, "\alpha"'] & & \\
& X \times_S Y \arrow[r, "p_2"] \arrow[d, "p_1"] & Y \arrow[d, "g"] \\
& X \arrow[r, "f"] & S
\end{tikzcd}$$

Example 4.26. The category of sets has all fibred products: $X \times_S Y \cong \{(x, y) \in X \times Y \mid f(x) = g(y)\}$. Some special situations: If $|S| = 1$, then $f(x) = g(y)$ is automatically satisfied, so $X \times_S Y = X \times Y$. If f and g are inclusions, then $X \times_S Y \cong X \cap Y$. If f is arbitrary and $g : \{s\} \hookrightarrow S$, then $X \times_S Y \cong f^{-1}(s)$.